

WRRRA M Bridge Research 0.5-0.8 r1

Reviewed Clock, Measurement and Feedback Collection

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1. Evaluation and executed scope

Evaluation	Evidence in this review
Verified/frozen inputs	SI conversion, downstream electron energy/mode, finite upstream address state/internal H and reviewed r1 ledger. Population, species, reservoir, clock and spatial coefficients remain disclosed construction inputs.
WRRRA transformation	Address-conditioned state -> same internal H -> nonduplicated SI ledger -> homogeneous pressure/expansion; adopted proper-time measurement/record and a separate effective spatial feedback path.
Outputs	Seven stages replayed: 1,116 stage checks + 188 review checks = 1,304 passed. Eight result JSON files are byte-identical in a clean directory.
Falsification	Reject an adapter if trace, positivity, energy, pressure, phase, current, convergence or input dependence contradicts its specification. A full covariant bridge remains open because its joint dynamics is not implemented.

Stage check counts

Stage	Execution	Passed
0.1	Typed scope contract	15
0.2	Address-conditioned CQ state	95
0.3	SI exchange without duplication	134
0.4	Homogeneous pressure/FRW	600
0.5	Conditional SI clock/phase	134
0.6	Measurement/record/repetition	87
0.7	Effective nonuniform feedback	51

Counts include mathematical/implementation checks, provenance and confirmation of expected limitations. They are not numbers of independent experiments or observational confirmations.

2. Executed connections and retained provenance

Connection	Status	Payload / limit
0.2 -> 0.3 -> 0.4	Executed	Excitation -> externally matched SI population load -> pressure/FRW from one E(V). Particle creation and preparation time law remain open.
0.2 + 0.5 -> 0.6	Executed	Same internal gap and adopted frame -> post-observation states/repeated transitions. Autonomous apparatus/reset remain open.
0.2 -> 0.7	Executed	Baseline populations -> chosen spatial profiles -> occupancy -> field -> state. Dimensionless nonrelativistic effective model.
0.5 -> 0.7	Provenance	Clock snapshot is retained. Changing its rate leaves field results unchanged. SI length/kinetic/G calibration remains open.
0.6 -> 0.7	Provenance	Measurement snapshot is retained; post-measurement state/apparatus/record energy is not fed into field evolution.
0.6 -> 0.3/0.4	Review diagnostic	0.8 inserts one mean write cost into the existing SI ledger and checks external/included supply boundaries. Actual apparatus space-time dynamics remains open.

Behavioral input-dependency probes

Isolated input change	Numerical response
Double 0.7 clock rate	Density/field/convergence unchanged; maximum density difference 0
Double 0.7 measurement work	Density/field/convergence unchanged; maximum density difference 0
Set 0.7 baseline population to 0.12	Maximum final density difference 0.01681125416
Double 0.6 frame duration	First repeated-transition matrix maximum difference 0.55243420946

These synthetic perturbations test actual code dependencies. The population perturbation is not counted as a recalibrated upstream state or a particle-creation calculation.

The 0.7 time-refinement ratio is 4.000041; the 64/128 grid density L2 difference is 9.02348e-12. This establishes convergence for the declared smooth finite test, not universal accuracy.

3. SI ledger check with measurement cost

Verified/frozen inputs: baseline excited population $p=0.07997665450$, the same-H gap Delta, 0.5 frame/0.6 record gap, and 0.3 fixed expected population $N_{\text{bar}}/\text{reservoir}$. The reservoir choice 0.2 E_{phi} is a diagnostic construction input.

WRRRA transformation: the chosen X readout changes the nonselective internal state to $1/2$. Mean one-write supply is $W=(1/2-p) \Delta_J + \text{record_gap_J}/2$. Internal excitation and record energy are each counted once. Total supply is $N_{\text{bar}} W$, opposed by reservoir change $-N_{\text{bar}} W$.

Outputs: inherited/proton calibration at reference 1 m3 gives excitation increase $7.0374395222\text{e-}12$ J, record increase $4.3176187869\text{e-}16$ J and total supply $7.0378712840\text{e-}12$ J. Reservoir energy falls from $7.5615824991\text{e-}12$ J to positive $5.2371121503\text{e-}13$ J.

Calibration/species	Supply / J	External q: before -> after	Included q
inherited/proton	$7.0378713\text{e-}12$	-0.528550 -> -0.519197	-0.518508
inherited/neutron	$7.0285264\text{e-}12$	-0.528550 -> -0.519209	-0.518508
alternate/proton	$7.1378005\text{e-}12$	-0.523000 -> -0.513566	-0.512871
alternate/neutron	$7.1283229\text{e-}12$	-0.523000 -> -0.513579	-0.512871

External boundary: the subsystem excluding its supplier gains the supplied energy. Included boundary: with the existing pressureless-supplier assumption, total energy and q are unchanged by redistribution. The two boundaries have different reference total energy, so their distinct q values are consistent.

Falsification: reject this ledger insertion if internal plus record increase differs from supply, the reservoir turns negative, included total energy changes, or pressure disagrees with the same $E(V)$. All four declared combinations pass.

Scope: a conditional one-write mean ledger followed by the fixed-comoving-energy rule. It does not implement $Q(t)$, actual apparatus distribution, record reset, 100,000 physical memory costs or nonuniform covariant gravity.

4. Consistency and resolution of apparent conflicts

Apparent conflict	Assessment / resolution
31.8% versus 100%	Resident Actual is retained phi+D; the complete ledger includes returns. Separate them through typed scope.
5/26.8/68.2 versus 4.93/26.5/68.57	Address probabilities and SI energy fractions are different input types. Use a named energy calibration rather than automatic substitution.
$q=-0.52855$ versus -0.523	The same pressure law reproduces each stated energy calibration. Always retain calibration and subsystem boundary.
Post-readout $p=0.5$ versus fixed prepared state in 0.4	Readout changes the state and requires work. A post-write $E(V)$ can use the new load; $Q(t)$ remains open.
Uniform $\phi=0$ in 0.7 versus FRW expansion	Periodic Poisson subtracts the mean and does not calculate the background zero mode. Do not count the uniform test as FRW reproduction; a common background/perturbation adapter remains open.
Clock/record snapshots imply field integration?	Perturbation confirms these files are not numerical dependencies of 0.7. Claim integration only for executed connections.

Reproduction, explanation and prediction

Reproducing known values through WRRR transformations is a consistency and explanatory achievement. The structural contribution is the executed connection of address-conditioned state, internal H , signed SI exchange, pressure and observation/record costs through typed ledgers and explicit boundaries.

After verified inputs and construction choices are fixed, repeated-transition probabilities, noise-conditioned record errors and finite spatial density/field evolution are conditional WRRR model predictions. At $p_{\text{flip}}=0.001$ and 256 ticks the predicted error is 0.2005056913. Observations must match the declared medium/noise conditions; a disagreement under matched conditions requires revising or rejecting that model rule.

5. Final status of the ten original bridge issues

Issue	Final status
B01	Scope distinction resolved
B02	Address/energy types and conditional calibration resolved; abundance origin open
B03	q reproduced for each stated calibration/boundary
B04	Diagonal-address CQ correlation and observable reduction executed; no general entanglement claim
B05	Nonduplication and signed finite supply executed; physical preparation/creation apparatus open
B06	Old clock equality remains failed; selected SI adapter executes, physical rate open
B07	Ideal instrument/work/record noise executed; physical durability/reset open
B08	Internal 95D, carrier 128D and channel spaces separated; carrier dynamics open
B09	Effective nonuniform feedback executes; full SI covariant bridge open
B10	Source-history correlations tested; actual species population/boundary are construction inputs

6. Verdict, replay and handoff to 1.0

Bridge development closes at the agreed 0.8. Preserve the finite conditional achievements of 0.1-0.7 and retain autonomous/covariant interfaces as open. Upstream closure at 0.10 and downstream at 0.12 remain unchanged. Version 1.0 consolidates this reviewed scope for publication; it does not erase open issues. Corrections use r1/r2 without automatic extra stages.

Replay: install numpy/scipy and run `OPENBLAS_NUM_THREADS=1 python reproduce_all.py`. Forward freshly generated state/basis and results in dependency order. Mark 0.7 clock/measurement copies as provenance. Eight result JSON files are byte-identical; input hashes are checked. See `environment.json`, `results.json` and `clean_replay.json`.

선행 공개 자료 / Prior public sources:

Bridge 0.1-0.4 r1: doi.org/10.5281/zenodo.23119802

Upstream 1.0-r1: doi.org/10.5281/zenodo.23119041

Downstream 0.10-0.12: doi.org/10.5281/zenodo.23113101

GitHub: github.com/Wonsik-Choi-janefather/wrra-m-0.1

Reviewed release DOI: [10.5281/zenodo.23120693](https://doi.org/10.5281/zenodo.23120693). Repository directory: `bridge/review_0_5_0_8_r1`. No journal submission or acceptance is claimed.

7. Revision r1 and release specification

Stage	Revision	Checks: original -> r1
0.5	Use a relative norm for the small SI generator Hermitian test; directly compare electron-mode calibration and SI energy scale.	133 -> 134
0.6	Verify the full 95D H Hermiticity, invariant two-mode support, projected relative generator and initial excitation energy.	83 -> 87
0.7	Check FFT kinetic Hermiticity, matrix/gradient energy equivalence and quantitative second-order density/energy refinement. Define drho as the discrete cell-mass derivative.	40 -> 51
0.8	Read original ZIP result payloads and compare all seven stages; verify ten issues/B09 open scope; propagate current outputs and freeze reviewed references.	179 -> 188

All scientific numerical payloads remain identical. Version/check/dependency metadata is intentionally updated. Original 0.5-0.8 ZIPs are retained in historical/. Reviewed 0.1-0.4 code is unchanged and bundled as a dependency.

The complete run passes 1,116 stage checks + 188 review checks = 1,304, an increase of 25 over the original 1,279. Revised 0.5-0.7 stages contribute 272 checks; prior stages contribute 844 dependency checks. The 188 review checks include the prior 36 cross-stage checks once.

Release files are Korean/English PDFs and the complete reproduction ZIP. The ZIP includes revised code, declared inputs, results, before/after comparison, final issue register, environment and SHA256SUMS. Preserve reproduction/explanation/conditional prediction under stated assumptions and retain autonomous apparatus/full SI covariance as open.

DOI: <https://doi.org/10.5281/zenodo.23120693>

GitHub: https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/tree/main/bridge/review_0_5_0_8_r1